Chapter test

41 Techniques in modern biotechnology

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following is ***not*** an application of biotechnology?

   A. production of wine with yeasts
   B. selective breeding of desired crops
   C. primary treatment of sewage
   D. production of vaccines with weakened pathogens ☐

2. Which of the following can be used as a vector in recombinant DNA technology?

   (1) plasmids
   (2) bacterial chromosomes
   (3) viruses

   A. (2) only
   B. (1) and (2) only
   C. (1) and (3) only
   D. (1), (2) and (3) ☐

★ 3. Which of the following combinations correctly shows the uses of restriction enzymes and DNA ligase in recombinant DNA technology?

	restriction enzyme	***DNA ligase***	
A.	joins DNA fragments	cuts DNA fragments	
B.	cuts DNA fragments	joins DNA fragments	
C.	transfers DNA fragments into cells	synthesizes new DNA strands	
D.	cuts DNA fragments	synthesizes new DNA strands	☐

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4. Which of the following combinations is/are ***incorrect***?

	Step in PCR	Process occurred
(1)	Primer annealing	Primers bind to the target sequence
(2)	Denaturation	Synthesis of new DNA strands
(3)	Extension	The DNA molecule separates into two separate strands

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

★ 5. If it takes 2 minutes to complete each cycle of PCR, how many copies of DNA can be produced in 16 minutes from a segment of DNA?

A. 32
B. 64
C. 256
D. 512 ☐

6. Which of the following is ***not*** an application of PCR?

A. forensic investigations
B. archaeological studies
C. blood group tests
D. assisting the diagnosis of infectious diseases ☐

7. Which of the following descriptions about the principle of DNA fingerprinting using VNTRs is ***incorrect***?

A. VNTRs are found in different loci of the coding DNA.
B. The number of repetitive base sequences in a VNTR varies among individuals.
C. Depending on the number of repeats in a VNTR, VNTRs of different lengths are observed in different individuals.
D. DNA fragments containing VNTRs can be extracted with restriction enzymes. ☐

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8. Which of the following shows the correct sequence of steps in RFLP analysis?

(1) Transfer the DNA fragments to the nylon membrane and incubate with radioactive probes.
(2) Cut the DNA sample with restriction enzymes.
(3) Place an X-ray film against the nylon membrane.
(4) Separate the DNA fragments with gel electrophoresis.
(5) Denature the double-stranded DNA into single strands.

A. (4), (5), (2), (1), (3)
B. (2), (4), (5), (1), (3)
C. (2), (5), (4), (3), (1)
D. (5), (2), (4), (1), (3) ☐

9. To create a transgenic animal by microinjection, the desired gene is injected into

A. a fertilized egg.
B. an unfertilized egg.
C. an embryo.
D. the uterus of the surrogate mother. ☐

★ 10. The genetic make-up of a cloned animal using somatic cell nuclear transfer would be identical to

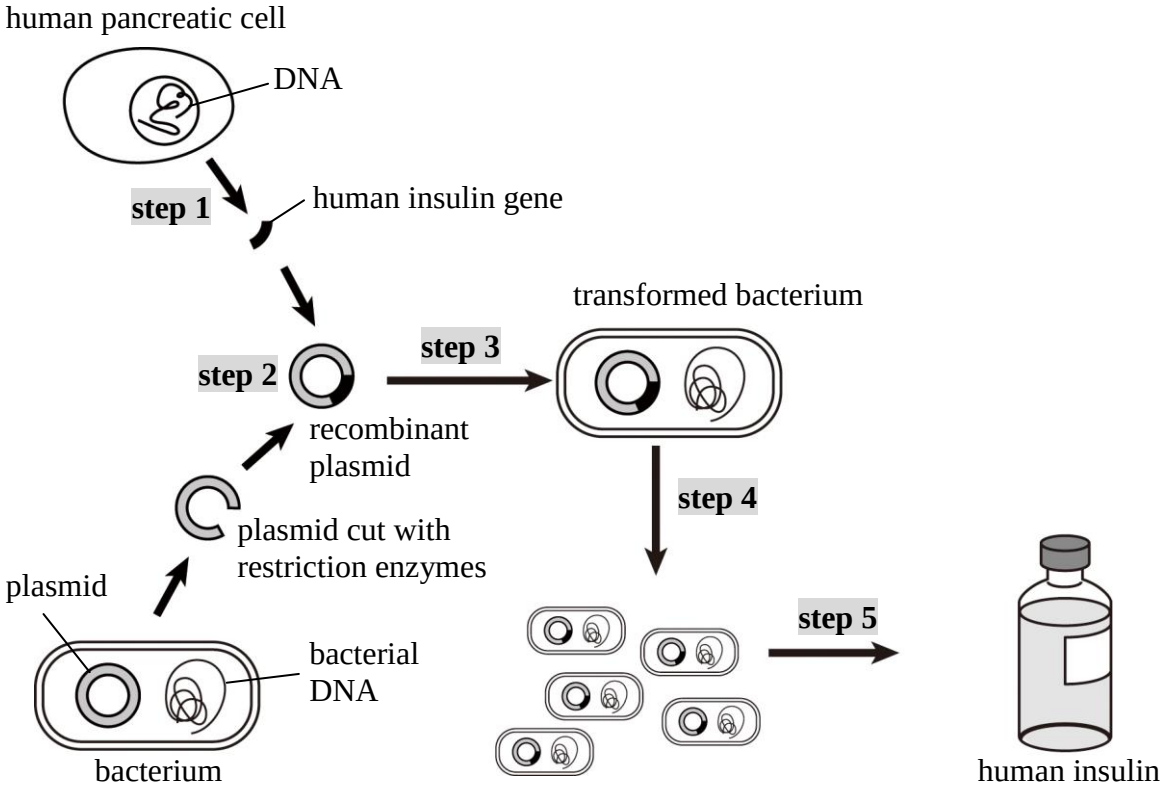
A. the egg cell donor.
B. the sperm cell donor.
C. the surrogate mother.
D. the somatic cell donor. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	41.1	41.2	41.2	41.2	41.2	41.2	41.2	41.2	41.2	41.3

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B. Problem solving (20 marks)

11. The diagram below outlines how human insulin is produced by genetically modified bacteria.



(a) Describe the ***five*** major steps involved in the production of genetically modified bacteria for manufacturing human insulin. (5 marks)

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(b) Describe and explain ***one*** method to select the transformed bacteria. (4 marks)

(Total: 9 marks)

★ 12. Golden Rice has been genetically modified to carry foreign genes which enhance its nutritional content. Its grains are orange in colour. Scientists claimed that Golden Rice can be used as a food-based approach to treat or prevent vitamin A deficiency, which is a serious public health problem affecting millions of children and pregnant women globally.

(a) Suggest ***one*** disease caused by vitamin A deficiency. (1 mark)

(b) (i) Describe how a foreign gene can be introduced into a plant cell. (5 marks)

(ii) Explain the use of antibiotic resistance genes in gene transfer. (2 marks)

(iii) Suggest ***one*** potential hazard of using antibiotic resistance genes in genetic engineering. (1 mark)

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(c) Vitamin A deficiency can cause serious eye disorders in children. Suggest ***two*** reasons why supplying vitamin A to children in the form of tablets might be better than Golden Rice. (2 marks)

(Total: 11 marks)

C. Exam challenge (10 marks)

★★ 13. Sickle-cell anaemia is caused by a single base mutation in the gene coding for haemoglobin. Using restriction enzyme X, PCR and DNA fingerprinting, doctors can detect the presence of the allele causing sickle-cell anaemia. X recognizes and cuts at the middle of the allele for normal haemoglobin (H^A), but it does not recognize the allele for sickle-cell haemoglobin (H^S).

The diagnosis of sickle-cell anaemia involves the following steps:

1. Extract DNA from the cells in the sample.
2. Carry out PCR using primers that are specific to part of the haemoglobin gene.
3. Add restriction enzyme X to the PCR product.
4. Carry out gel electrophoresis to separate the DNA fragments according to their molecular size.

(a) What is a primer? (1 mark)

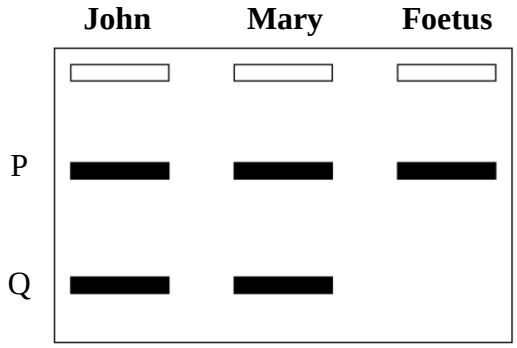
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(b) Other than primers, suggest ***one*** raw material of PCR. (1 mark)

(c) Explain why it is necessary to carry out PCR in step 2. (1 mark)

(d) John and Mary are both carriers of sickle-cell anaemia. Mary is pregnant and the couple wants to know whether their baby will have sickle-cell anaemia.

A doctor used the method described above to test the DNA samples from John, Mary and the foetus. The PCR products consist of DNA fragments with the length of 110 base pairs. After treatment with restriction enzyme X, the resulted DNA fingerprints are as follows.



(i) Explain John's band pattern. (5 marks)

(ii) Deduce whether the foetus has sickle cell anaemia or not. (2 marks)

(Total: 10 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	C	2.	C	3.	B	4.	D	5.	C
6.	C	7.	A	8.	B	9.	A	10.	D

(1 mark each)

B. Problem solving (20 marks)

11. (a) Step 1: The gene coding for human insulin is isolated from the human chromosome. (1)
Step 2: The human insulin gene is inserted into a cut plasmid to form a recombinant plasmid. (1)
Step 3: The recombinant plasmid is introduced into a bacterium. (1)
Step 4: The transformed bacteria are cultured to large quantities. (1)
Step 5: Human insulin is extracted from the bacterial culture and purified. (1)

(b) Use antibiotic resistance gene as selection marker. (1)
Antibiotic resistance gene is inserted into the plasmid. (1)
The bacteria that have been mixed with the recombinant plasmids are spread onto an agar plate which contains the antibiotic. (1)
The transformed bacteria can survive and form colonies while the rest cannot. (1)

(Total: 9 marks)

12. (a) night blindness (or other acceptable answers) (1)

(b) (i) The foreign genes and T_i plasmids are cut by the same restriction enzyme. (1)
The foreign genes and T_i plasmids are joined by DNA ligase to form recombinant plasmids. (1)
The recombinant plasmids are introduced into *Agrobacteria*. (1)
The transformed *Agrobacteria* are used to infect plant cells. (1)
The recombinant plasmids carry the foreign genes are integrated into the plant DNA. (1)

(ii) They act as selection markers which allow identification and selection of the transformed cells. (1)
The markers give the transformed cells characteristics that allow them to survive in an environment where the non-transformed cells cannot. (1)

(iii) The genes may pass to pathogenic bacteria. Theses pathogens may become resistant to antibiotics / become ‘superbug’. (1)

(c) The dosage of vitamin A tablets is more precise. (1) /
People who do not have rice as their main food cannot benefit from Golden Rice. (1) /

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The transport / storage / distribution of vitamin A tablets is more convenient. (1) / Children may be feeding mainly on milk but not rice. (1)
(or other acceptable answers) (Any ***two***, 2)

(Total: 11 marks)

C. Exam challenge (10 marks)

13. (a) A short sequence of single-stranded DNA which is complementary to a specific region on a DNA template (1)

(b) free nucleotides (do not accept DNA polymerase) (1)

(c) To produce enough copies of DNA for analysis (1)

(d) (i) John has both the allele for normal haemoglobin (H^A) and the allele for sickle-cell haemoglobin (H^S). (1)
The alleles for normal haemoglobin (H^A) are recognized and cut by restriction enzyme X into halves. (1)
These shorter fragments move faster during electrophoresis and they correspond to the band Q. (1)
The alleles for sickle-cell haemoglobin (H^S) are not cut by X. They remain long fragments. (1)
The longer fragments move slower during electrophoresis and they correspond to the band P. (1)

(ii) The foetus has sickle-cell anaemia / is homozygous for sickle-cell anaemia. (1)
Its DNA fingerprint consists of band Q only, which indicates the presence of the alleles for sickle-cell haemoglobin (H^S) only. (1)

(Total: 10 marks)

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